Control of an Articulated-Vehicle System Comprising Dual Multi-Axle Trailers

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***Abstract*— This paper presents a novel modeling approach of an articulated-vehicle system comprising dual multi-axle trailers which does not belong to conventional trailer systems such as multi-steering n-trailer and multi-steering general n-trailer systems. In the new model, some existing mechanical elements are replaced with virtual counterparts, and furthermore, virtual mechanical elements are incorporated, facilitating the conversion of the kinematical equation of the articulated-vehicle system into a chained form. This paper also presents a control law, derived from the chained form, for enabling four manipulation points defined on the articulated-vehicle system to follow a single desired free-form curve path. The validity of the modeling approach and the control law is verified by experiments.**

I. Introduction

This paper presents a path-following feedback control law for an articulated-vehicle system comprising dual multi-axle trailers, specially designed for transportation of super-large, super-heavy objects as shown in Fig.1.

In general, a trailer system constitutes a combination in which a powered vehicle (tractor) is articulated to a trailer via a passive joint. A multi-trailer system constitutes a combination in which a tractor is articulated via a passive joint to multiple trailers all of which are interconnected in series via passive joints. The multi-trailer systems which encompass the trailer systems are categorized into the two groups, depending on the structural feature of their connection points: (a) the n-trailer systems; and (b) the general n-trailer systems.

The structure of the n-trailer systems [1] [2] is noteworthy due to the fact that all their connection points are on-axle hitchings, each having a passive joint positioned either on the tractor rear axle or on the trailer axle. The structural feature of on-axle hitchings allows the conversion of the kinematical equations of the n-trailer systems into a canonical form, e.g., a chained form [1] [2]. A control method of the n-trailer systems, derived from the chained form, for point stabilization and path-following has been proposed [2]. The structure of the general n-trailer systems [3] [4] is also noteworthy due to the fact that all their connection points are either on-axle or off-axle hitchings. An off-axle hitching has a passive joint positioned either off the tractor rear axle or off the trailer axle. The structural feature of on-axle and off-axle hitchings allows the conversion of the kinematical equations of the general n-trailer systems into the chained form by incorporating a virtual steering system at each off-axle hitching [3] . The angle of each of the virtual steering systems is not independent and determined by the state variables of the general n-trailer systems. Controllability of them has been proven [3] and a model predictive control method for the path-following motion has been proposed [4].

In these two types of the trailer systems, a significant characteristic is that there are only two control inputs: the velocity of the tractor and the angular velocity of its steering. Especially, the motion of the trailers depends on and is uniquely determined by the motion of the tractor. There is no degree of freedom to independently control each trailer to avoid interference with its neighboring obstacles. This means that maneuverability of the n-trailer and general n-trailer systems is not sufficient.

The multi-steering n-trailer systems [5] [6] have been introduced, all of whose connection points are on-axle hitchings, allowing the conversion of their kinematical equations into the chained form, in which the degree of freedom of control increases with the number of the steering systems, i.e., which has more than two control inputs. The greater the number of the steering systems, the higher the maneuverability. A control method of the multi-steering n-trailer systems, derived from the chained form, for steering using polynomial inputs has been proposed [5]. As examples of the multi-steering n-trailer systems, a cooperative transportation system with two car-like mobile robots [7] [8] and a five-axle three-steering articulated-vehicle system [9] [10] have been introduced and their control methods, derived from the chained form, have been proposed. Utilizing higher maneuverability, two manipulation points defined on the cooperative transportation system are guided to independently follow two desired free-form curve paths, with the quantitative specification of the motion of its carrier determined by the shapes of these paths [8]. The two carriers of the five-axle three-steering articulated-vehicle system are capable of forming straight and V-shaped configurations, depending on the shapes of objects transported [9] [10]. The validity of these systems and their control methods have been verified by experiments [7] [8] [9] [10]. The multi-steering

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general n-trailer systems [11] [12] have also been introduced. Higher maneuverability of them has been proven [11] and a control method for off-track reduction in path-following motion along roundabouts has been proposed [12].

The articulated-vehicle system discussed in this paper comprises dual multi-axle trailers, each towed by a tractor. In the multi-axle trailer, the load is distributed across its multiple wheels, facilitating the transport of heavier objects. To prevent any slippage in the wheels, the extension lines of their axles are required to be either parallel or intersect at a single point. Satisfying these conditions, the multi-axles are modeled as a single axle, as shown in Fig.2. As understood from Fig.2, a carrier onto which objects are loaded has no wheel installations, so that one of its joints is regarded as neither an on-axle hitching nor an off-axle hitching. In other words, the articulated-vehicle system in Fig.1 is categorized into neither the group of the multi-steering n-trailer systems nor that of the multi-steering general n-trailer systems. It is not straightforward to design a control law for the articulated-vehicle system in Fig.1.

This paper presents a novel modeling approach of the articulated-vehicle system in which some of existing mechanical elements are replaced with virtual counterparts, and furthermore, virtual mechanical elements are incorporated. The kinematical equation of the new model is converted into the chained form. It is important to note that not all the state variables of the new model are included in the chained form: one state variable depends on others. This is significantly different from the conventional trailer systems. A feedback control law, derived from the chained form, is proposed which enables four manipulation points defined on the articulated-vehicle system to follow a single desired free-form curve path. The validity of the modeling approach and the control law is verified by experiments.

II. Structure of Articulated-Vehicle System

The basic structure of the articulated-vehicle system comprising dual multi-axle trailers each of which is towed by an individual vehicle is shown in Fig.1. Vehicle-1 is articulated to Multi-axle Trailer-1 via Joint-1 which is passive. Multi-axle Trailer-1 has a large number of passive wheels all of which are equipped with steering functionality, thus enabling active control of the rolling direction of the passive wheels relative to its orientation. Multi-axle Trailer-1 is articulated to Object via Joint-2 which is passive. It is feasible to model Object by replacing it with a carrier onto which it is loaded. Object is articulated to Multi-axle Trailer-2 via Joint-3 which is passive. Multi-axle Trailer-2 is articulated to Vehicle-2 via Joint-4 which is passive. Vehicle-1 and Multi-axle Trailer-1 to which it is articulated form a tractor-trailer system with an off-axle hitching in which Joint-1 is not positioned on the rear axle of Vehicle-1. Similarly, Vehicle-2 and Multi-axle Trailer-2 form another. Especially, Object (which can be modeled as a carrier) is equipped with no wheels and it is articulated to the dual tractor-trailer systems, which means that the articulated-vehicle system in Fig.1 does not belong to the category of the multi-steering general n-trailer systems all of whose hitchings are either on-axle or off-axle. Therefore, the control laws for the multi-steering general n-trailer systems are not directly applicable to the articulated-vehicle system in Fig.1.

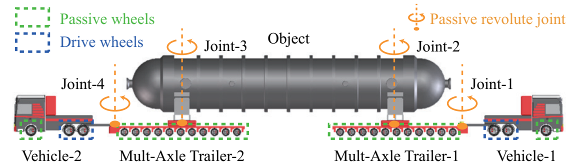


Fig. 1. An articulated-vehicle system comprising dual multi-axle trailers each of which is towed by an individual vehicle (tractor)

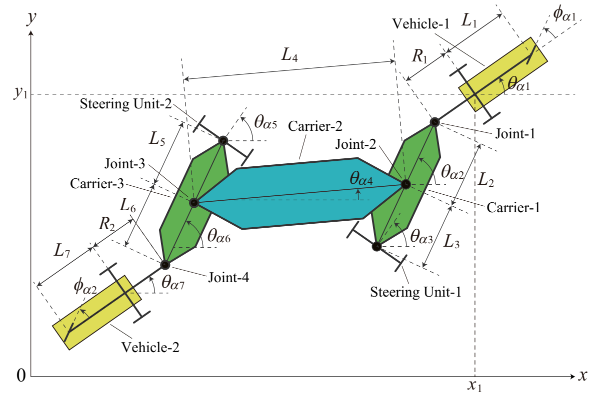


Fig. 2. An original model of an articulated-vehicle system

III. Original Model of Articulated-Vehicle System

It is known that, in order to prevent any slippage in the wheels of Multi-axle Trailer-1 in Fig.1, the extension lines of all the wheel axles are required to satisfy either of the conditions: (i) being parallel to each other or (ii) intersecting each other at a single point. In these conditions, all the axles of Multi-axle Trailer-1 can be modeled as a single axle as shown in Fig.2. Specifically, all the steering systems of Multi-axle Trailer-1 are modeled as Steering Unit-1. Similarly, those of Multi-axle Trailer-2 are modeled as Steering Unit-2. Multi-axle Trailer-1 is replaced with Carrier-1 with Steering Unit-1 as well as Multi-axle Trailer-2 is replaced with Carrier-3 with Steering Unit-2 in Fig.2. Object which is replaced with Carrier-2 articulated to Carriers-1 and -3 via Joints-2 and -3, respectively. Additionally, the two front steering wheels of Vehicle-1 are modeled as a single steering wheel, similar to how those of Vehicle-2 are modeled as a single one. Such model in Fig.2 is referred to as an ”original model” which is kinematically equivalent to the articulated-vehicle system in Fig.1.

IV. Kinematical Equation of Original Model in Cartesian Coordinate System

The state variable vector $\boldsymbol{X}$ of the original model of the articulated-vehicle system in Fig.2 in the Cartesian coordinate system is defined as:

$$\boldsymbol{X} = (x_1, y_1, \theta_{\alpha 1}, \phi_{\alpha 1}, \theta_{\alpha 2}, \theta_{\alpha 3}, \theta_{\alpha 4}, \theta_{\alpha 5}, \theta_{\alpha 6}, \theta_{\alpha 7}, \phi_{\alpha 2})^T, \tag{1}$$

where x_1 and y_1 represent the position of the midpoint of the Vehicle-1 rear axle along the x-axis as well as the y-axis; $\theta_{\alpha 1}$ represents the orientation of Vehicle-1; $\phi_{\alpha 1}$ the steering angle of Vehicle-1; $\theta_{\alpha 2}$ the orientation of Carrier-1; $\theta_{\alpha 3}$ the orientation of Steering Unit-1; $\theta_{\alpha 4}$ the orientation of Carrier-2; $\theta_{\alpha 5}$ the orientation of Steering Unit-2; $\theta_{\alpha 6}$ the orientation of Carrier-3; $\theta_{\alpha 7}$ the orientation of Vehicle-2; and $\phi_{\alpha 2}$ the steering angle of Vehicle-2.

The kinematical equation of the original model of the articulated-vehicle system is given as:

$$\dot{\boldsymbol{X}} = \sum_{i=1}^{5} \boldsymbol{G}_i v_i, \tag{2}$$

$$\begin{cases} \boldsymbol{G_1} = (G_{11}, G_{12}, \cdots, G_{111})^T, \\ \boldsymbol{G_2} = (0,0,0,1,0,0,0,0,0,0,0)^T, \\ \boldsymbol{G_3} = (0,0,0,0,0,1,0,0,0,0,0)^T, \\ \boldsymbol{G_4} = (0,0,0,0,0,0,0,1,0,0,0)^T, \\ \boldsymbol{G_5} = (0,0,0,0,0,0,0,0,0,0,1)^T, \end{cases} \tag{3}$$

$$\begin{cases} G_{11} = \cos\theta_{\alpha 1}, \\ G_{12} = \sin\theta_{\alpha 1}, \\ G_{13} = \dfrac{\tan\phi_{\alpha 1}}{L_1}, \\ G_{14} = 0, \\ G_{15} = \dfrac{\sin\theta_{\alpha 1,3} - R_1 G_{13}\cos\theta_{\alpha 1,3}}{L_{2,3}\cos\theta_{\alpha 2,3}}, \\ G_{16} = 0, \\ G_{17} = -\dfrac{\Lambda}{L_4}, \\ G_{18} = 0, \\ G_{19} = \dfrac{V_{\mathrm{rear}}\sin\theta_{\alpha 5,7} - R_2 G_{110}\cos\theta_{\alpha 5,7}}{L_{5,6}\cos\theta_{\alpha 5,6}}, \\ G_{110} = -\dfrac{V_{\mathrm{rear}}\tan\phi_{\alpha 2}}{L_7}, \\ G_{111} = 0, \end{cases} \tag{4}$$

$$\begin{cases} L_{i,j} = L_i + L_j, \\ \theta_{\alpha i,j} = \theta_{\alpha i} - \theta_{\alpha j}, \\ \Lambda = R_1 G_{13}\cos\theta_{\alpha 1,4} + L_2 G_{15}\cos\theta_{\alpha 2,4} \\ \quad + L_6 G_{19}\cos\theta_{\alpha 4,6} + R_2 G_{110}\cos\theta_{\alpha 4,7} \\ \quad - \sin\theta_{\alpha 1,4} - V_{\mathrm{rear}}\sin\theta_{\alpha 4,7}, \\ V_{\mathrm{rear}} = \dfrac{L_{5,6}L_7(\lambda_1 + \lambda_2)}{L_1 L_{2,3}(L_{5,6}\cos\theta_{\alpha 4,6}\lambda_3 - \sin\theta_{\alpha 4,6}\lambda_4)}, \\ \lambda_1 = \cos\theta_{\alpha 1,2}(L_1 L_{2,3}\cos\theta_{\alpha 2,4} \\ \quad + \sin\theta_{\alpha 2,4}(L_3 R_1 \tan\phi_{\alpha 1} + L_1 L_2 \tan\theta_{\alpha 2,3})), \\ \lambda_2 = \sin\theta_{\alpha 1,2}(L_{2,3} R_1 \cos\theta_{\alpha 2,4}\tan\phi_{\alpha 1} \\ \quad + \sin\theta_{\alpha 2,4}(-L_1 L_3 + L_2 R_1 \tan\phi_{\alpha 1}\tan\theta_{\alpha 2,3})), \\ \lambda_3 = L_7\cos\theta_{\alpha 6,7} - R_2\sin\theta_{\alpha 6,7}\tan\phi_{\alpha 2}, \\ \lambda_4 = \cos\theta_{\alpha 6,7}(L_5 R_2 \tan\phi_{\alpha 2} - L_6 L_7 \tan\theta_{\alpha 5,6}) \\ \quad + \sin\theta_{\alpha 6,7}(L_5 L_7 + L_6 R_2 \tan\phi_2 \tan\theta_{\alpha 5,6}), \end{cases} \tag{5}$$

where L_1 represents the distance between the front and rear axles of Vehicle-1; L_2 the distance between Joint-1 connecting Vehicle-1 to Carrier-1 and Joint-2 connecting Carrier-1 to Carrier-2; L_3 the distance between Joint-2 and the axis of Steering Unit-1 attached to Carrier-1; L_4 the distance between Joint-2 and Joint-3 connecting Carrier-2 to Carrier-3; L_5 the distance between Joint-3 and the axis of Steering Unit-2 attached to Carrier-3; L_6 the distance between Joint-3 and Joint-4 connecting Carrier-3 to Vehicle-2; L_7 the distance between the rear and front axles of Vehicle-2; R_1 the length of the off-hook structure between Vehicle-1 and Carrier-1; R_2 the length of the off-hook structure between Carrier-3 and Vehicle-2; v_1 the velocity of the midpoint of the Vehicle-1 rear axle; v_2 the angular velocity of the steering system of Vehicle-1; v_3 the angular velocity of Steering Unit-1; v_4 the angular velocity of Steering Unit-2; and v_5 the angular velocity of the steering system of Vehicle-2.

V. Control Model of Articulated-Vehicle System

As understood from Sections II and III, Vehicle-1 and Carrier-1 with Steering Unit-1 in Fig.2 form a tractor-trailer system with an off-axle hitching as well as Vehicle-2 and Carrier-3 with Steering Unit-2 form another. Unlike conventional trailers which are equipped with wheels, Carrier-2 has no wheel installations and it constraints the relative distance between the dual tractor-trailer systems, e.g., the relative distance between Joint-2 and Joint-3, so that the articulated-vehicle system in Fig.2 does not belong to the group of the multi-steering general n-trailer systems all of whose hitchings are either on-axle or off-axle. Therefore, the kinematical equation of the articulated-vehicle system in Fig.2 cannot be directly converted into a canonical form such as a chained form.

To utilize the control laws designed based on the chained form, some mechanical elements are replaced with virtual mechanical elements, and additionally, new virtual mechanical elements are incorporated as shown in Fig.3. Specifically, Carrier-1 in Fig.2 is divided into Virtual Carriers-1 and 2 which are articulated via a virtual joint installed at Joint-2 in Fig.3. Similarly, Carrier-3 in Fig.2 is divided into Virtual Carriers-3 and -4 articulated via another virtual joint installed at Joint-3 in Fig.3. Additionally, Virtual Steering Units-1, -2, -3 and -4 are installed at Joints-1, -2, -3 and -4, respectively. Especially, Vehicle-1 has three axles, e.g., those of a front steering wheel, rear wheels and Steering Unit-1 wheels, so that, depending on the orientation of the three axles, their wheels may impede each other's motion, resulting in an overconstraint. To prevent such overconstraint, the steering angle of the front wheel of Vehicle-1 is determined by the orientation of Vehicle-1 and Steering Unit-1 accordingly. Therefore, the front steering wheel of Vehicle-1 is excluded, as well as that of Vehicle-2 as shown in Fig.3. Such model in Fig.3 is referred to as a "control model." For ensuring the equivalence of the control model to the original model, Virtual Carriers-1 and -2 are required to be aligned with each other, as are Virtual Carriers-3 and -4.

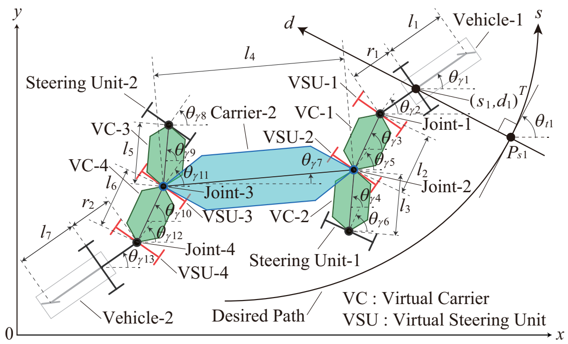


Fig. 3. A control model of an articulated-vehicle system

VI. Path-Following Motion

This paper presents a new path-following feedback control law which causes the four manipulation points $\boldsymbol{P_1}$ (the midpoint of the Vehicle-1 rear axle), $\boldsymbol{P_2}$ (Joint-2), $\boldsymbol{P_3}$ (Joint-3) and $\boldsymbol{P_4}$ (the midpoint of the Vehicle-2 rear axle) defined on the articulated-vehicle system to follow a single desired free-form curve path, respectively, as shown in Fig.4. Of course, in order to ensure the equivalence of the control model in Fig.3 to the original model in Fig.2, it is necessary to constantly maintain Virtual Carrier-1 aligned with Virtual Carrier-2 as well as Virtual Carrier-3 with Virtual Carrier-4.

When the midpoint of the Vehicle-1 rear axle is off the desired path, a point $\boldsymbol{P_{s1}}$ on the path which determines the position of the midpoint relative to the path is defined as follows. As shown in Fig.3, the point $\boldsymbol{P_{s1}}$ satisfies the condition of the perpendicular intersection between the tangent of the path at $\boldsymbol{P_{s1}}$ and the straight line through $\boldsymbol{P_{s1}}$ to the midpoint. The position of $\boldsymbol{P_{s1}}$ along the s-axis defined on the path is denoted by s_1. The position of the midpoint of the Vehicle-1 rear axle relative to $\boldsymbol{P_{s1}}$ along the d-axis defined on the straight line through $\boldsymbol{P_{s1}}$ to the midpoint is denoted by d_1.

Especially, as shown in Fig.4, $\boldsymbol{P_{s1}}$ is the target point of $\boldsymbol{P_1}$. A point $\boldsymbol{P_{s2}}$ is the target point of $\boldsymbol{P_2}$ and it is defined as follows. $\boldsymbol{P_{s2}}$ is one of the intersections between the path and a circle whose center is located at Joint-1 and whose radius is the length of Virtual Carrier-1 and it is closest to $\boldsymbol{P_2}$, when $\boldsymbol{P_1}$ coincides with $\boldsymbol{P_{s1}}$ and the orientation of Vehicle-1 aligns with the direction of the tangent of the path at $\boldsymbol{P_{s1}}$. A point $\boldsymbol{P_{s3}}$ is the target point of $\boldsymbol{P_3}$ and it is one of the intersections between the path and a circle whose center is located at Joint-2 and whose radius is the length of Carrier-2 and it is closest to $\boldsymbol{P_3}$, when $\boldsymbol{P_2}$ coincides with $\boldsymbol{P_{s2}}$. A point $\boldsymbol{P_{s4}}$ is the target point of $\boldsymbol{P_4}$ and it is one of the points on the path and is located at a distance of the second off-hook length from Joint-4 which is in the direction of the tangent of the path at $\boldsymbol{P_{s4}}$, and it is closest to $\boldsymbol{P_4}$, when $\boldsymbol{P_3}$ coincides with $\boldsymbol{P_{s3}}$. The position of $\boldsymbol{P_{s2}}$, $\boldsymbol{P_{s3}}$ and $\boldsymbol{P_{s4}}$ along the s-axis is denoted by s_2, s_3 and s_4, respectively. The direction of the tangent of the path at $\boldsymbol{P_{s2}}$, $\boldsymbol{P_{s3}}$ and $\boldsymbol{P_{s4}}$ is denoted by θ_{t2}, θ_{t3} and θ_{t4}, respectively.

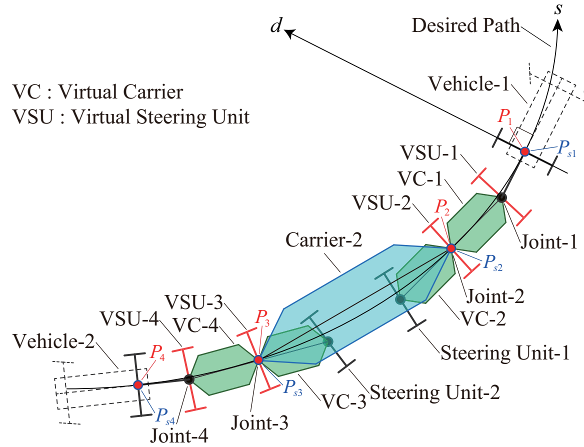


Fig. 4. Path-following motion of an articulated-vehicle system in a control model

VII. Kinematical Equation of Control Model in Curvilinear Coordinate System

The state variable vector $\boldsymbol{x}$ of the control model of the articulated-vehicle system in Fig.3 in the Cartesian coordinate system is given as:

$$\begin{aligned}\boldsymbol{x} = (&x_1, y_1, \theta_{\gamma 1}, \theta_{\gamma 2}, \theta_{\gamma 3}, \theta_{\gamma 4}, \theta_{\gamma 5}, \theta_{\gamma 6},\\ &\theta_{\gamma 7}, \theta_{\gamma 8}, \theta_{\gamma 9}, \theta_{\gamma 10}, \theta_{\gamma 11}, \theta_{\gamma 12}, \theta_{\gamma 13})^T,\end{aligned} \tag{6}$$

where x_1 and y_1 represent the position of the midpoint of the Vehicle-1 rear axle along the x-axis as well as y-axis; $\theta_{\gamma 1}$ represents the orientation of Vehicle-1; $\theta_{\gamma 2}$ the orientation of Virtual Steering Unit-1; $\theta_{\gamma 3}$ the orientation of Virtual Carrier-1; $\theta_{\gamma 4}$ the orientation of Virtual Carrier-2; $\theta_{\gamma 5}$ the orientation of Virtual Steering Unit-2; $\theta_{\gamma 6}$ the orientation of Steering Unit-1; $\theta_{\gamma 7}$ the orientation of Carrier-2; $\theta_{\gamma 8}$ the orientation of Steering Unit-2; $\theta_{\gamma 9}$ the orientation of Virtual Carrier-3; $\theta_{\gamma 10}$ the orientation of Virtual Carrier-4; $\theta_{\gamma 11}$ the orientation of Virtual Steering Unit-3; $\theta_{\gamma 12}$ the orientation of Virtual Steering Unit-4; and $\theta_{\gamma 13}$ the orientation of Vehicle-2.

The state variable vector $\boldsymbol{x}_{sd}$ of the control model of the articulated-vehicle system in Fig.3 in the curvilinear coordinate system is given as:

$$\begin{aligned}\boldsymbol{x}_{sd} = (&s_1, d_1, \theta_{\gamma p1}, \theta_{\gamma p2}, \theta_{\gamma p3}, \theta_{\gamma p4}, \theta_{\gamma p5}, \theta_{\gamma p6},\\ &\theta_{\gamma p7}, \theta_{\gamma p8}, \theta_{\gamma p9}, \theta_{\gamma p10}, \theta_{\gamma p11}, \theta_{\gamma p12}, \theta_{\gamma p13})^T,\end{aligned} \tag{7}$$

$$\begin{cases} \theta_{\gamma pi} = \theta_{\gamma i} - \theta_{t1}, \\ i = 1, 2, \cdots, 13, \end{cases} \tag{8}$$

where s_1 and d_1 represent the position of the midpoint of the Vehicle-1 rear axle along the s-axis as well as the d-axis in the curvilinear coordinate system; and θ_{t1} represents the direction of the tangent of the desired path at $\boldsymbol{P_{s1}}$, as shown in Fig.3.

The kinematical equation of the control model of the articulated-vehicle system is given as:

$$\dot{\boldsymbol{x}}_{sd} = \sum_{i=1}^{7} \boldsymbol{g}_{sdi} u_i, \tag{9}$$

$$\begin{cases} \boldsymbol{g}_{sd1} = (g_{sd11}, g_{sd12}, \cdots, g_{sd115})^T, \\ \boldsymbol{g}_{sd2} = (0,0,0,1,0,0,0,0,0,0,0,0,0,0,0)^T, \\ \boldsymbol{g}_{sd3} = (0,0,0,0,0,1,0,0,0,0,0,0,0,0,0)^T, \\ \boldsymbol{g}_{sd4} = (0,0,0,0,0,0,1,0,0,0,0,0,0,0,0)^T, \\ \boldsymbol{g}_{sd5} = (0,0,0,0,0,0,0,0,1,0,0,0,0,0,0)^T, \\ \boldsymbol{g}_{sd6} = (0,0,0,0,0,0,0,0,0,0,0,1,0,0,0)^T, \\ \boldsymbol{g}_{sd7} = (0,0,0,0,0,0,0,0,0,0,0,0,0,1,0)^T. \end{cases} \tag{10}$$

$$\begin{cases} g_{sd11} = \dfrac{\cos\theta_{\gamma p1}}{1 - d_1 c(s_1)}, \\ g_{sd12} = \sin\theta_{\gamma p1}, \\ g_{sd13} = \dfrac{\tan\theta_{\gamma p1,2}}{r_1} - \Gamma, \\ g_{sd14} = -\Gamma, \\ g_{sd15} = \dfrac{\sin\theta_{\gamma p2,5}}{l_2 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5}} - \Gamma, \\ g_{sd16} = \dfrac{\cos\theta_{\gamma p2,3} \sin\theta_{\gamma p5,6}}{l_3 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5} \cos\theta_{\gamma p4,6}} - \Gamma, \\ g_{sd17} = -\Gamma, \\ g_{sd18} = -\Gamma, \\ g_{sd19} = -\dfrac{\cos\theta_{\gamma p2,3} \sin\theta_{\gamma p11,5}}{l_4 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5} \cos\theta_{\gamma p11,7}} - \Gamma, \\ g_{sd110} = -\Gamma, \\ g_{sd111} = -\dfrac{\cos\theta_{\gamma p2,3} \cos\theta_{\gamma p5,7} \sin\theta_{\gamma p11,8}}{l_5 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5} \cos\theta_{\gamma p11,7} \cos\theta_{\gamma p8,9}} \\ \quad -\Gamma, \\ g_{sd112} = \dfrac{\cos\theta_{\gamma p2,3} \cos\theta_{\gamma p5,7} \sin\theta_{\gamma p11,12}}{l_6 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5} \cos\theta_{\gamma p11,7} \cos\theta_{\gamma p10,12}} \\ \quad -\Gamma, \\ g_{sd113} = -\Gamma, \\ g_{sd114} = -\Gamma, \\ g_{sd115} = \dfrac{\cos\theta_{\gamma p2,3} \cos\theta_{\gamma p5,7} \cos\theta_{\gamma p10,11} \sin\theta_{\gamma p12,13}}{r_2 \cos\theta_{\gamma p1,2} \cos\theta_{\gamma p3,5} \cos\theta_{\gamma p11,7} \cos\theta_{\gamma p10,12}} \\ \quad -\Gamma, \end{cases} \tag{11}$$

$$\theta_{\gamma pi,j} = \theta_{\gamma pi} - \theta_{\gamma pj}, \tag{12}$$

$$\Gamma = \frac{c(s_1)\cos\theta_{\gamma p1}}{1 - d_1 c(s_1)}, \tag{13}$$

$$\begin{cases} l_i = L_i, \ i = 1, 2, \cdots, 7, \\ r_i = R_i, \ i = 1, 2, \end{cases} \tag{14}$$

where u_1 represents the moving velocity of the midpoint of the Vehicle-1 rear axle; u_2 the angular velocity of Virtual Steering Unit-1; u_3 the angular velocity of Virtual Steering Unit-2; u_4 the angular velocity of Steering Unit-1; u_5 the angular velocity of Steering Unit-2; u_6 the angular velocity of Virtual Steering Unit-3; u_7 the angular velocity of Virtual Steering Unit-4; and $c(s_1)$ is the curvature of the desired path at $\boldsymbol{P_{s1}}$.

VIII. Conversion into Chained Form

The kinematical equation, Eq.(9), of the control model of the articulated-vehicle system in the curvilinear coordinate system is converted into the time differential equation in a chained form as follows. First, the seven vector fields $\boldsymbol{g}_{sd1}$, $\boldsymbol{g}_{sd2}$, $\cdots$, $\boldsymbol{g}_{sd7}$ in Eq.(9) are converted as:

$$\begin{cases} \boldsymbol{f}_{sd1} = \boldsymbol{g}_{sd1} \dfrac{1 - d_1 c(s_1)}{\cos\theta_{\gamma p1}}, \\ \boldsymbol{f}_{sdi} = \boldsymbol{g}_{sdi}, \ i = 2, 3, \cdots, 7. \end{cases} \tag{15}$$

Secondly, the seven control inputs u_1, u_2, $\cdots$, u_7 in Eq.(9) are converted as:

$$\begin{cases} \tilde{u}_1 = u_1 \dfrac{\cos\theta_{\gamma p1}}{1 - d_1 c(s_1)}, \\ \tilde{u}_i = u_i, \ i = 2, 3, \cdots, 7. \end{cases} \tag{16}$$

Then, Eq(9) is rewritten as:

$$\dot{\boldsymbol{s}}_{sd} = \sum_{i=1}^{7} \boldsymbol{f}_{sdi} \tilde{u}_i. \tag{17}$$

By using the seven vector fields $\boldsymbol{f}_{sd1}$, $\boldsymbol{f}_{sd2}$, $\cdots$, $\boldsymbol{f}_{sd7}$, the state variables of $\boldsymbol{x}_{sd}$ except $\theta_{\gamma p13}$ in Eq.(7) are converted as:

$$\begin{cases} z_{11} = h_1 = s_1, \\ z_{21} = L^2_{\boldsymbol{f}_{sd1}} h_2, \\ z_{22} = L_{\boldsymbol{f}_{sd1}} h_2, \\ z_{23} = h_2 = d_1, \\ z_{31} = L_{\boldsymbol{f}_{sd1}} h_3, \\ z_{32} = h_3 = \theta_{\gamma p3}, \\ z_{41} = L_{\boldsymbol{f}_{sd1}} h_4, \\ z_{42} = h_4 = \theta_{\gamma p4}, \end{cases} \quad \begin{cases} z_{51} = L_{\boldsymbol{f}_{sd1}} h_5, \\ z_{52} = h_5 = \theta_{\gamma p7}, \\ z_{61} = L_{\boldsymbol{f}_{sd1}} h_6, \\ z_{62} = h_6 = \theta_{\gamma p9}, \\ z_{71} = L_{\boldsymbol{f}_{sd1}} h_7, \\ z_{72} = h_7 = \theta_{\gamma p10}. \end{cases} \tag{18}$$

Here, $L_{\boldsymbol{p}} q$ is a Lie derivative of a scalar q along a vector field $\boldsymbol{p}$. Thus, the time derivatives of the variables in Eq.(18) become the time differential equations in a six-chain, single-generator chained form as:

$$\begin{cases} \dot{z}_{11} = w_1, \\ \dot{z}_{21} = w_2, \\ \dot{z}_{22} = z_{21} w_1, \\ \dot{z}_{23} = z_{22} w_1, \end{cases} \quad \begin{cases} \dot{z}_{i1} = w_i, \\ \dot{z}_{i2} = z_{i1} w_1, \\ i = 3, 4, \cdots, 7, \end{cases} \tag{19}$$

$$\begin{cases} w_1 = \tilde{u}_1, \\ w_2 = \displaystyle\sum_{j=1}^{7} L_{\boldsymbol{f}_{sdj}} L^2_{\boldsymbol{f}_{sd1}} h_2 \tilde{u}_j, \\ w_i = \displaystyle\sum_{j=1}^{7} L_{\boldsymbol{f}_{sdj}} L_{\boldsymbol{f}_{sd1}} h_i \tilde{u}_j, \\ i = 3, 4, \cdots, 7. \end{cases} \tag{20}$$

It is important to note that the state variable $\theta_{\gamma p13}$ in Eq.(7) which represents the orientation of Vehicle-2 is not incorporated into the time differential equations in the chained form in Eq.(19). The variation of $\theta_{\gamma p13}$ is determined by those of the other state variables in Eq.(7) accordingly. In this aspect, the articulated-vehicle system is significantly different from other articulated-vehicle systems in which all their state variables are incorporated into time differential equations in the chained form.

To derive the control law which allows variability of the moving velocity of the midpoint of the Vehicle-1 rear axle along the s-axis, a new derivative is introduced as:

$$\begin{cases} z_{2i}^{(1)} = \mathrm{sign}(w_1)\dfrac{\partial z_{2i}}{\partial z_{11}}, \\ z_{jk}^{(1)} = \mathrm{sign}(w_1)\dfrac{\partial z_{jk}}{\partial z_{11}}, \\ i=1,2,3,\ j=3,4,\cdots,7,\ k=1,2. \end{cases} \tag{21}$$

Eq.(21) represents the derivative with respect to a variable $\mathrm{sign}(w_1)z_{11}$ obtained by multiplying the sign function $\mathrm{sign}(w_1)(=\mathrm{sign}(\dot{z}_{11}))$ and the position $z_{11}(=s_1)$ of the midpoint of the Vehicle-1 rear axle along the s-axis, instead of time t. Since $\mathrm{sign}(w_1)\Delta z_{11}(=\mathrm{sign}(w_1)w_1\Delta t)>0$, $w_1\neq 0$, as $\Delta t>0$, the velocities of change of the state variables of $\boldsymbol{x}_{sd}$ in Eq.(7) except s_1, in the differential equation with respect to $\mathrm{sign}(w_1)z_{11}$, depends on the velocity of change of s_1, e.g., the moving velocity w_1 of the midpoint of the Vehicle-1 rear axle along the s-axis, so that the velocity of change of $\boldsymbol{x}_{sd}$ is automatically adjusted by w_1 and the variability of w_1 is allowed without adjustment of control parameters such as feedback gains.

Eqs.(19) and (20) are rewritten as:

$$\begin{cases} z_{11}^{(1)}=\mathrm{sign}(w_1), \\ z_{21}^{(1)}=\mathrm{sign}(w_1)\tilde{w}_2, \\ z_{22}^{(1)}=\mathrm{sign}(w_1)z_{21}, \\ z_{23}^{(1)}=\mathrm{sign}(w_1)z_{22}, \end{cases} \quad \begin{cases} z_{i1}^{(1)}=\mathrm{sign}(w_1)\tilde{w}_i, \\ z_{i2}^{(1)}=\mathrm{sign}(w_1)z_{i1}, \\ i=3,4,\cdots,7. \end{cases} \tag{22}$$

$$\begin{cases} w_i = w_1\tilde{w}_i, \\ i=2,3,\cdots,7. \end{cases} \tag{23}$$

IX. Control Law

This paper presents a new control law for achieving the path-following motion in Fig.4. In the control law, the moving velocity w_1 of the midpoint of the Vehicle-1 rear axle along the s-axis is designed as:

$$w_1 = f_v(t), \tag{24}$$

where $f_v(t)$ is a time-varying function which can be close to zero and can even be zero.

In the control law, the position $d_1(=z_{23})$ of the midpoint of the Vehicle-1 rear axle relative to the desired path along the d-axis is caused to converge to its desired value $d_{1d}(=z_{23d})$. For the convergence of d_1 to d_{1d}, a Lyapunov function V_{21} which represents the square of the deviation of z_{23} from z_{23d} is defined as:

$$V_{21} = (z_{23d}-z_{23})^2, \tag{25}$$

$$z_{23d} = d_{1d}. \tag{26}$$

Based on the Lyapunov second method, in order to cause V_{21} to converge to zero, the control input w_2 is designed as:

$$w_2 = w_1\tilde{w}_2 = w_1\mathrm{sign}(w_1)z_{23}^{(3)}, \tag{27}$$

$$\begin{aligned} z_{23}^{(3)} &= z_{23d}^{(3)} + k_{21}\left(z_{23d}^{(2)}-z_{23}^{(2)}\right) \\ &\quad +k_{22}\left\{z_{23d}^{(2)}+k_{21}\left(z_{23d}^{(1)}-z_{23}^{(1)}\right)-z_{23}^{(2)}\right\} \\ &\quad +k_{23}\Big[z_{23d}^{(2)}+k_{21}\left(z_{23d}^{(1)}-z_{23}^{(1)}\right) \\ &\quad +k_{22}\left\{z_{23d}^{(1)}+k_{21}\left(z_{23d}-z_{23}\right)-z_{23}^{(1)}\right\}-z_{23}^{(2)}\Big]. \end{aligned} \tag{28}$$

The orientation $\theta_{\gamma p3}(=z_{32})$ of Virtual Carrier-1 in the curvilinear coordinate system is caused to converge to $\theta_{\gamma p3d}$ as well as the orientation of $\theta_{\gamma p4}(=z_{42})$ of Virtual Carrier-2 to $\theta_{\gamma p4d}$. Especially, $\theta_{\gamma p3d}$ is set to be identical to $\theta_{\gamma p4d}$, satisfying that Virtual Carriers-1 and -2 form a combined carrier which is straight and which is kinematically equivalent to Carrier-1 in the original model in Fig.2. The orientation $\theta_{\gamma p7}$ $(=z_{52})$ of Carrier-2 is caused to converge to $\theta_{\gamma p7d}$. Similarly, the orientation $\theta_{\gamma p9}$ $(=z_{62})$ of Virtual Carrier-3 is caused to converge to $\theta_{\gamma p9d}$ as well as the orientation $\theta_{\gamma p10}$ $(=z_{72})$ of Virtual Carrier-4 to $\theta_{\gamma p10d}$, and $\theta_{\gamma p9d}$ is set to be identical to $\theta_{\gamma p10d}$, satisfying that Virtual Carriers-3 and -4 form a straight combined carrier which is kinematically equivalent to Carrier-3 in the original model in Fig.2. For the convergence of the variables z_{i2}, $i=3,4,\cdots,7$ to z_{i2d}, $i=3,4,\cdots,7$, the Lyapunov functions V_{i1}, $i=3,4,\cdots,7$, which represent the squares of the deviations of z_{i2}, $i=3,4,\cdots,7$, from z_{i2d}, $i=3,4,\cdots,7$, as:

$$\begin{cases} V_{i1} = (z_{i2d}-z_{i2})^2, \\ i=3,4,\cdots,7, \end{cases} \tag{29}$$

$$\begin{cases} z_{32d}=\theta_{\gamma p3d}, \\ z_{42d}=\theta_{\gamma p4d}, \\ z_{52d}=\theta_{\gamma p7d}, \\ z_{62d}=\theta_{\gamma p9d}, \\ z_{72d}=\theta_{\gamma p10d}. \end{cases} \tag{30}$$

Based on the Lyapunov second method, in order to cause V_{i1}, $i=3,4,\cdots,7$, to converge to zero, the control inputs w_i, $i=3,4,\cdots,7$, are designed as:

$$\begin{cases} w_i = w_1\tilde{w}_i = w_1 z_{i2}^{(2)}, \\ i=3,4,\cdots,7. \end{cases} \tag{31}$$

$$\begin{cases} z_{i2}^{(2)} = z_{i2d}^{(2)} + k_{i1}\left(z_{i2d}^{(1)}-z_{i2}^{(1)}\right) \\ \qquad +k_{i2}\left\{z_{i2d}^{(1)}+k_{i1}\left(z_{i2d}-z_{i2}\right)-z_{i2}^{(1)}\right\}, \\ i=3,4,\cdots,7. \end{cases} \tag{32}$$

For achieving the path-following motion in Fig.4, the midpoint of the Vehicle-1 rear axle, Joints-2, -3 and the midpoint of the Vehicle-2 rear axle are required to follow the single desired path. Also, for ensuring the equivalence of the control model of the articulated-vehicle system to its original model, Virtual Carrier-1 is required to be constantly aligned with Virtual Carrier-2 as well as Virtual Carrier-3 with Virtual Carrier-4. For both the path-following motion and the equivalence of the control model to the original model, the desired values, d_{1d} of the position of the midpoint of the Vehicle-1 rear axle relative to the desired path, $\theta_{\gamma p3d}$ of the orientation of Virtual Carrier-1 in the curvilinear coordinate system, $\theta_{\gamma p4d}$ of the orientation of Virtual Carrier-2, $\theta_{\gamma p7d}$ of the orientation of Carrier-2, $\theta_{\gamma p9d}$ of the orientation of Virtual Carrier-3 and $\theta_{\gamma p10d}$ of the orientation of Virtual Carrier-4 are designed as:

$$\begin{cases} d_{1d} = 0, \\ \theta_{\gamma p3d} = \text{atan2}\,(P_{sy}(s_1) - r_1 \sin\theta_{t1} - P_{sy}(s_2), \\ \qquad\qquad P_{sx}(s_1) - r_1 \cos\theta_{t1} - P_{sx}(s_2)) - \theta_{t1}, \\ \theta_{\gamma p4d} = \theta_{\gamma p3d}, \\ \theta_{\gamma p7d} = \text{atan2}\,(P_{sy}(s_2) - P_{sy}(s_3), \\ \qquad\qquad P_{sx}(s_2) - P_{sx}(s_3)) - \theta_{t1}, \\ \theta_{\gamma p9d} = \theta_{\gamma p10d}, \\ \theta_{\gamma p10d} = \text{atan2}\,(P_{sy}(s_3) - P_{sy}(s_4) - r_2 \sin\theta_{t4}, \\ \qquad\qquad P_{sx}(s_3) - P_{sx}(s_4) - r_2 \cos\theta_{t4}) - \theta_{t1}, \end{cases} \tag{33}$$

where $(P_{sx}(s), P_{sy}(s))^T$ represents a position vector of a point on the desired path.

X. Conversion of Control Inputs

As mentioned in Section III, the kinematical model of the articulated-vehicle system in Fig.1 is equivalent to the original model in Fig.2. However, in the control model in Fig.3, (i) Carrier-1 is divided into Virtual Carriers-1 and -2 as well as Carrier-2 into Virtual Carriers-3 and -4; (ii) Virtual Steering Units-1, -2, -3 and -4 are newly defined; and (iii) the front steering wheels of Vehicles-1 and -2 are excluded, so that the control inputs u_i, $i = 1, 2, \cdots, 7$, in Eq.(9) of the control model in Fig.3 are required to be converted into the control inputs v_i, $i = 1, 2, \cdots, 5$, in Eq.(2) of the original model in Fig.2, satisfying the kinematical constraints in the original model, as follows.

The steering angle $\phi_{\alpha 1}$ of Vehicle-1 in the original model is determined by the orientation $\theta_{\gamma 1}$ of Vehicle-1 and the orientation $\theta_{\gamma 2}$ of Virtual Steering Unit-1 in the control model accordingly, as well as the steering angle $\phi_{\alpha 2}$ of Vehicle-2 in the original model is determined by the orientation $\theta_{\gamma 12}$ of Virtual Steering Unit-4 and the orientation $\theta_{\gamma 13}$ of Vehicle-2 in the control model accordingly, as:

$$\begin{cases} \phi_{\alpha 1} = \tan^{-1}\left(-\dfrac{l_1 \tan(\theta_{\gamma 2} - \theta_{\gamma 1})}{r_1}\right), \\ \phi_{\alpha 2} = \tan^{-1}\left(-\dfrac{l_7 \tan(\theta_{\gamma 12} - \theta_{\gamma 13})}{r_2}\right). \end{cases} \tag{34}$$

The control inputs v_i, $i = 1, 2, \cdots, 5$, in Eq.(2) of the original model into which the control inputs u_i, $i = 1, 2, \cdots, 7$, in Eq.(9) of the control model are converted, are given as:

$$\begin{cases} v_1 = u_1, \\ v_2 = \dot{\phi}_{\alpha 1} \\ \quad = \dfrac{l_1 r_1 (g_{13} u_1 - u_2)}{\cos^2(\theta_{\gamma 1} - \theta_{\gamma 2})\left(r_1^2 + l_1^2 \tan^2(\theta_{\gamma 1} - \theta_{\gamma 2})\right)}, \\ v_3 = u_4, \\ v_4 = u_5, \\ v_5 = \dot{\phi}_{\alpha 2} \\ \quad = \dfrac{l_7 r_2 (g_{115} u_1 - u_7)}{\cos^2(\theta_{\gamma 12} - \theta_{\gamma 13})\left(r_2^2 + l_7^2 \tan^2(\theta_{\gamma 12} - \theta_{\gamma 13})\right)}, \end{cases} \tag{35}$$

$$\begin{cases} \theta_{\gamma 1} = \theta_{\alpha 1}, \\ \theta_{\gamma 2} = \theta_{\alpha 1} - \text{atan2}\,(R_1 \tan\phi_{\alpha 1}, L_1), \\ \theta_{\gamma 3} = \theta_{\alpha 2}, \\ \theta_{\gamma 4} = \theta_{\alpha 2}, \\ \theta_{\gamma 5} = \theta_{\alpha 2} - \text{atan2}\,(L_2 \tan(\theta_{\alpha 2} - \theta_{\alpha 3}) \\ \qquad\qquad + L_3 \tan(\theta_{\alpha 2} - \theta_{\gamma 2}), L_2 + L_3), \\ \theta_{\gamma 6} = \theta_{\alpha 3}, \\ \theta_{\gamma 7} = \theta_{\alpha 4}, \\ \theta_{\gamma 8} = \theta_{\alpha 5}, \\ \theta_{\gamma 9} = \theta_{\alpha 6}, \\ \theta_{\gamma 10} = \theta_{\alpha 6}, \\ \theta_{\gamma 11} = \theta_{\alpha 6} - \text{atan2}\,(L_5 \tan(\theta_{\alpha 6} - \theta_{\alpha 5}) \\ \qquad\qquad + L_6 \tan(\theta_{\alpha 6} - \theta_{\gamma 12}), L_5 + L_6), \\ \theta_{\gamma 12} = \theta_{\alpha 7} - \text{atan2}\,(R_2 \tan\phi_{\alpha 2}, L_7), \\ \theta_{\gamma 13} = \theta_{\alpha 7}. \end{cases} \tag{36}$$

XI. Experimental Results

To verify the validity of the modeling approach and the control law, an experimental apparatus of the articulated-vehicle system shown in Figs.5 and 6 has been developed.

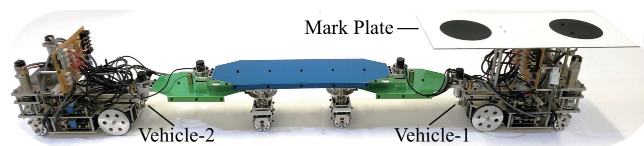


Fig. 5. An experimental apparatus of an articulated-vehicle system

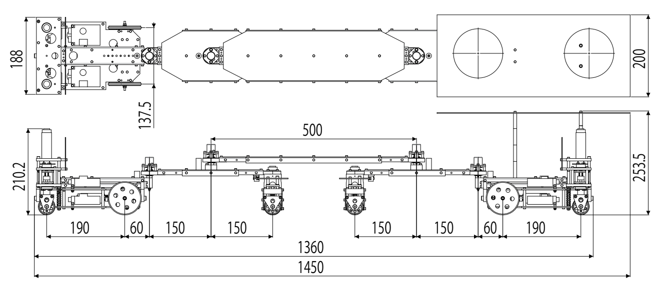


Fig. 6. Dimensions of an articulated-vehicle system

The parameters which determine the size of the articulated-vehicle system are given as:

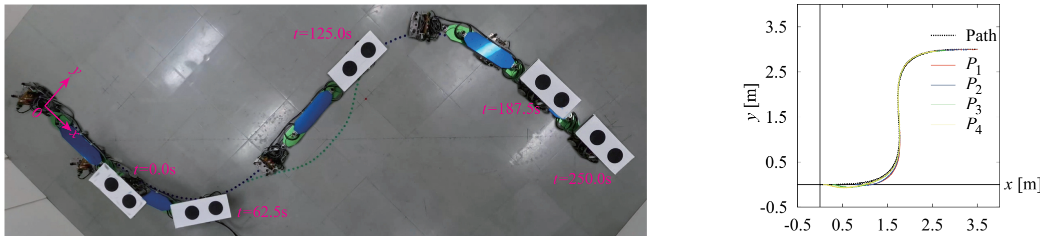
$$\begin{cases} L_1 = 0.190\text{m},\ L_i = 0.150\text{m},\ i = 2, 3, \\ L_4 = 0.500\text{m},\ L_j = 0.150\text{m},\ j = 5, 6, \\ L_7 = 0.190\text{m},\ R_k = 0.060\text{m},\ k = 1, 2. \end{cases} \tag{37}$$

The six control points of a fifth-order Bezier curve which is a single desired path are arranged as:

$$\begin{cases} \boldsymbol{P_0} = (0.0\text{m}, 0.0\text{m})^T, & \boldsymbol{P_1} = (2.0\text{m}, 0.0\text{m})^T, \\ \boldsymbol{P_2} = (2.5\text{m}, 0.0\text{m})^T, & \boldsymbol{P_3} = (1.0\text{m}, 3.0\text{m})^T, \\ \boldsymbol{P_4} = (1.5\text{m}, 3.0\text{m})^T, & \boldsymbol{P_5} = (3.5\text{m}, 3.0\text{m})^T. \end{cases} \tag{38}$$

The moving velocity w_1 of the midpoint of the Vehicle-1 rear axle along the s-axis is designed as:

$$w_1 = f_{vd}(t) + k_{w_1}(s_{1d}(t) - z_{11}), \tag{39}$$



(a) Snapshots at 62.5-second intervals (b) Trajectories of $\boldsymbol{P_i}$, $i = 1, 2, 3, 4$

Fig. 7. An experimental result

$$f_{vd}(t) = \begin{cases} k_1 t, \ 0 \leq t \leq T_1, \\ k_2, \ T_1 < t \leq T_2, \\ k_2 - k_1 t, \ T_2 < t \leq T_3, \end{cases} \tag{40}$$

$$s_{1d}(t) = \int_0^t f_{vd}(t)dt + s_{1d}(0), \tag{41}$$

$$\begin{cases} k_1 = 0.000327\mathrm{m/s}^2, \\ k_2 = 0.03\mathrm{m/s}, \end{cases} \tag{42}$$

$$\begin{cases} T_1 = 91.8\mathrm{s}, \\ T_2 = 158.2\mathrm{s}, \\ T_3 = 250.0\mathrm{s}, \end{cases} \tag{43}$$

$$k_{w_1} = 1.0. \tag{44}$$

The control parameters in Eqs.(28) and (32) are designed as:

$$\begin{cases} k_{2i} = 4.0, \ i = 1, 2, 3, \\ k_{jk} = 8.0, \ j = 3, 4, \cdots, 7, \ k = 1, 2. \end{cases} \tag{45}$$

The initial conditions of the position and orientation of the articulated-vehicle system are given as:

$$\begin{cases} (x_1, y_1)^T|_{t=0} = (1.0\mathrm{m}, 0.0\mathrm{m})^T, \\ \theta_{\alpha i}|_{t=0} = 0.0\mathrm{rad}, \ i = 1, 2, \cdots, 7, \\ \phi_{\alpha j}|_{t=0} = 0.0\mathrm{rad}, \ j = 1, 2. \end{cases} \tag{46}$$

An experimental result is shown in Fig.7(a),(b). It is understood from Fig.7(a),(b) that the four manipulation points, the midpoint of the Vehicle-1 rear axle, Joints-2, -3 and the midpoint of the Vehicle-2 rear axle, rigidly follow the single desired path which is the fifth-order Bezier curve path, respectively. As a consequence, the validity of the modeling approach and the control law has been verified by the experiments (see Video [13] for detailed demonstration).

XII. Conclusions

This paper has presented a novel modeling approach of an articulated-vehicle system comprising dual multi-axle trailers which does not belong to conventional trailer systems such as multi-steering n-trailer and multi-steering general n-trailer systems, and a control law for path-following motion. In the new model, some existing mechanical elements are replaced with virtual counterpatrs, and furthermore, virtual mechanical elements are incorporated, facilitating the conversion of the kinematical equation of the articulated-vehicle system into a canonical form, e.g., a chained form. Unlike the conventional trailer systems, not all the state variables of the articulated-vehicle system are included in the chained form, so that it is not straightforward to design its control system. The control law, derived from the chained form, enables four manipulation points defined on the articulated-vehicle system to follow a single desired free form curve path, specifying its motion quantitatively by the shape of the path. The validity of the modeling approach and the control law has been verified by experiments.

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